# George Panagopoulos

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## Experience

- **University of Luxembourg**, Postdoctoral Scientist, Luxembourg (01.2024 - Current)
  - Developed **active learning** methods to facilitate genomic experiments where interventions are budgeted.
  - Researching **causal effect prediction** methods for efficient advertising experiments.
  - Implementing **equivariant neural networks** to predict interatomic forces for material discovery.
  - Coordinated and contributed to competitive research proposals with multi-stakeholder teams (academia and industry).
- **Amazon**, Applied Scientist, Luxembourg (04.2022 - 06.2023)
  - Led an applied research project revolving around **routing and scheduling**.
  - Developed methodologies based on **linear programming** to improve the transportation network's efficiency.
  - Managed multiple stakeholders from varying backgrounds (business, science, and project managers) to satisfy the dependencies and facilitate the projects' growth.
- **Ecole Polytechnique Informatics Laboratory (LIX)**, Research Engineer, Paris, France (06.2018 - 03.2022)
  - Conducted research on **spatiotemporal forecasting** using **graph neural networks** and **meta-learning** with applications in epidemic prediction and control.
  - Proposed novel solutions for **combinatorial optimization** problems using **mult-task, graph, and reinforcement learning**, with applications in social networks.
  - Combined **transformers** and **graph analytics** to analyze big data from social media, bibliographical databases, French legal documents and medical ontologies.
- **University of Houston, Computational Physiology Lab**, RA/TA, Houston, TX (08.2016 - 05.2018)
  - Developed a **forecasting** technique for driving anomalies using signals from wearable sensors on the driver, thermal imaging, and automotive sensors.
  - Conduct **statistical analysis** and **causal inference** on behavioral experiments to quantify the effect of stress on the performance of resident surgeons.
  - Developed **multi-task** learning for EEG signals from brain-computer interfaces experiments to reduce inter-subject variance.
- **National Center for Scientific Research Demokritos**, Research Assistant, Athens, Greece (08.2014 - 07.2016)
  - Designed and ran interactive experiments to using low-cost **brain computer interfaces** to build models that predict facial expressions using EEG.
  - Implemented web services that use **natural language processing** and unsupervised machine learning to model essays written by children, supporting three languages and facilitating Android games that enhance creativity.

## Education

- **Ph.D. in Computer Science**, Ecole Polytechnique, Paris, France (01.2019 - 03.2022)
  - Thesis Title: *"Learning Influence Representations: Methods and Applications"*
  - Supervisor: Prof. Michalis Vazirgiannis, Asst. Prof. Fragkiskos Malliaros
- **Ms. in Computer Science**, University of Houston, Houston, TX (08.2016 - 05.2018)
  - Thesis Title: *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"*
  - Supervisor: Prof. Ioannis Pavlidis
- **Bsc. in Informatics and Telematics**, Harokopio University of Athens, Greece (09.2010 - 07.2014)
  - Thesis Title: *"Detecting Rising Stars in Dynamic Collaborative Networks"*
  - Supervisor: Prof. Iraklis Varlamis

## Publications

1. G. Panagopoulos, J.F. Lutzeyer, S. Ennadir, M. Vazirgiannis, J. Pang *"Efficient Data Selection for Training Genomic Perturbation Models"* in arXiv, 2025

2. T. P. Tong, A. Wang, G. Panagopoulos and J. Pang *"Kolmogorov-Arnold Network for Gene Regulatory Network Inference"* to be presented in Computation Methods in Systems Biology (**CMSB**), 2025

3. R. Bresson, G. Nikolentzos, G. Panagopoulos, M. Chatzianastasis, J. Pang, M. Vazirgiannis. *"Kagnns: Kolmogorov-arnold networks meet graph learning"* to appear in Transaction on Machine Learning Research (**TMLR**), 2025

4. G. Panagopoulos, N. Tziortziotis, F. Malliaros, J. Pang, M. Vazirgiannis. *"Learning Graph Representations for Influence Maximization"* in Social Networks Analysis and Mining (**SNAM**), 2024

5. G. Panagopoulos, D. Malitesta, F. Malliaros, J. Pang. *"Uplift Modeling Under Limited Supervision"* European Conference on Machine Learning and Knowledge Discovery in Databases (**ECML PKDD**), 2024

6. G. Panagopoulos, N. Tziortziotis, F. Malliaros, M. Vazirgiannis. *"Maximizing influence with graph neural networks"* Advances in Social Networks Analysis and Mining (**ASONAM**), 2023

7. C. Kosma, G. Nikolentzos, G. Panagopoulos, J.M. Steyaert, M. Vazirgiannis. *"Neural Ordinary Differential Equations for Modeling Epidemic Spreading"*, in Transactions on Machine Learning Research (**TMLR**), 2023

8. G. Nikolentzos, G. Panagopoulos, M. Vazirgiannis. *"An Empirical Study of the Expressiveness of Graph Kernels and Graph Neural Networks"*, in International Conference on Artificial Neural Networks (**ICANN**), 2021

9. B. Rozemberczki , P. Scherer, Y. He, G. Panagopoulos, M. Astefanoaei, O. Kiss, F. Beres, N. Collignon, R. Sarkar. *"PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models"*. in Conference on Information and Knowledge Management (**CIKM resource track**), 2021

10. G. Panagopoulos, G. Nikolentzos, M. Vazirgiannis.*"Transfer Graph Neural Networks for Pandemic Forecasting"*, in AAAI Conference on Artificial intelligence (**AAAI**), 2021

11. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Multi-task Learning for Influence Estimation and Maximization"*, in **IEEE Transactions of Knowledge and Data Engineering**, 2020, 10.1109/TKDE.2020.3040028

12. G. Panagopoulos, H. Jalalzai *"Graph Neural Networks with Extreme Nodes Discrimination"*, in Deep Learning on Graphs: Methods and Applications, (**DLG@KDD**), 2020

13. P. Boniol, G. Panagopoulos, C. Xypolopoulos,R. E. Hamdani,D. R Amariles, M. Vazirgiannis. *"Performance in the courtroom: Automated processing and visualization of appeal court decisions in France"*, in Natural Legal Language Processing (**NLLP @ KDD**), 2020

14. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Influence Maximization using Influence and Susceptibility Embeddings"*, in AAAI International Conference on Web and Social Media (**ICWSM**), 2020

15. G. Panagopoulos, C. Xypolopoulos, K. Skianis, C. Giatsidis, J. Tang, M. Vazirgiannis. *"Scientometrics for Success and Influence in the Microsoft Academic Graph"*, in Complex Networks and Their Applications (**Complex Networks**), Lisbon, Portugal December 10-12, 2019

16. G. Panagopoulos, I. Pavlidis. *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"* **IEEE Transactions of Intelligent Transportation Systems** (2019)

17. I. Pavlidis, D. Zavlin, A. Khatri, A. Wesley, G. Panagopoulos, and A. Echo. *"Absence of Stressful Conditions Accelerates Dexterous Skill Acquisition in Surgery"* **Nature Scientific Reports** 9.1 (2019):1747

18. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"DiffuGreedy: An Influence Maximization Algorithm based on Diffusion Cascades"*, in Proc. of Complex Networks and Their Applications (**Complex Networks**), Cambridge, UK December 11-13, 2018

19. G. Panagopoulos. *"A Review of Network Inference Techniques for Neural Activation Time Series"*, Poster in Network Science (**NetSci**). Paris, France, June 11-15, 2018

20. G. Panagopoulos. *"Multi-Task Learning for Commercial Brain Computer Interfaces"*, in Proc. of IEEE BioInformatics and BioEngineering (**BIBE**), Washington DC, USA, October 23-25, 2017

21. G. Panagopoulos, G. Tsatsaronis, and I. Varlamis. *"Detecting Rising Stars in Dynamic Collaborative Networks"* **Elsevier Journal of Informetrics** 11.1 (2017): 198-222.

22. G. Panagopoulos, C. Palmer. *"A Specialized Interactive Data Application for EEG Based Sleep Studies"*, ADH Workshop in Pervasive Technologies Related to Assistive Environments (**PETRA**), Rhodes, Greece, June 21-23, 2017

23. G. Panagopoulos, P. Karampiperis, A. Koukourikos, S. Konstantinidis. *"Creativity Profiling Server: Modelling the Principal Components of Human Creativity over Texts"*, DECAT Workshop in User Modelling, Adaptation and Personalization (**UMAP**), Dublin, Ireland, June 19-July 3, 2015

24. P. Karampiperis, A. Koukourikos, G. Panagopoulos. *"From Computational Creativity Metrics to the Principal Components of Human Creativity"*, in Proc. of Knowledge, Information and Creativity Support Systems (**KICSS**), Limassol, Cyprus, November 6-8, 2014

25. A. Koukourikos, P. Karampiperis, G. Panagopoulos. *"Creative Stories: A Storytelling Game fostering Creativity"*, in Proc. of Cognition and Exploratory Learning in Digital Age (**CELDA**), Porto, Portugal, October 25-27, 2014

## Teaching Experience

1. Graph Machine Learning DP-CSCE Course, Belval, Luxembourg [TODO] (01.2026)
   - Designed and will give two 90 minute lectures on graph machine learning in graduate students of University of Luxembourg.
   - Designed and will assess 4 assignments (22 hours in total) on graph theory, graph neural networks, graph transformers and their applications.
2. Complex Network Analysis, Belval, Luxembourg (11.2024 - )
   - Designed/updated and gave three 3-hour lectures and tutorials in diffusion algorithms and graph neural networks, in the context of the 'Data Sicence MSc'.
   - Developed and supervised a novel data science competition for the course's main project.
3. Mentor in Learning on Graphs Summer School, Londong, United Kingdom (07.2024)
   - Developed a week-long project on causal machine learning.
   - Supervised closely 4 graduate students working on the project.
4. Tutor in Deepmind Mediterranean Machine Learning Summer School, Thessaloniki, Greece (07.2023)
   - Gave 5 tutorials on graduate-level students in different subfields of machine learning, including transformers, diffusion models etc.
   - Designed a 3-hour tutorial on graph neural networks in collaboration with scientists from Google.
5. Tutor in International Conference on Web Engineering, Biarritz, France (05.2021)
   - Gave a 3 hour tutorial on influence learning and maximization in collaboration with Prof. Fragkiskos Malliaros.
6. Teaching Assistant in Ecole Polytechnique Informatics Laboratory (LIX), Paris, France (06.2018 - 03.2022)
   - Gave lectures, designed and gave tutorials, developed data science competitions and graded assignments in graduate and undergraduate level courses:
     - Advanced Learning for Text and Graph Data
     - Machine Learning in Network Science
     - Virus Propagation
     - Artificial Intelligence Executive Education
7. Teaching Assistant in University of Houston, Houston, TX (08.2016 - 05.2018)
   - Gave lectures, designed and gave tutorials, held office hours and graded assignments in Statistical Methods in Research.
   - Held office hours and graded assignments in Software Engineering.
8. Tutor National Center for Scientific Research Demokritos, Athens, Greece (08.2014 - 07.2016)
   - Designed and gave weekly interactive lectures and tutorials for brain computer interfaces in undergraduate students.

## Co-supervision

- Tsz Pan Tong, University of Luxembourg (2026).
- Chrysoula Kosma, Ecole Polytechnique (2024).
- Iakovos Evdaimon (Msc), Athens University of Economics and Business (2021).

## Awards & Scholarships

- CIKM Best Resource Paper, 2021
- ICWSM Best paper honorable mention, 2020
- Registration on Machine Learning Summer School, 2020
- KDD Registration Award, 2020
- Best Ms Thesis Award, University of Houston, 2018
- PETRA Travel Award, 2017
- Presidential Fellowship Award, University of Houston, 2016
- PhD Fellowship, University of Houston, 2016
- Doctoral Scholarship, NCSR Demokritos, 2016
- 1st place in Open Public Data Hackathon, Greek Ministry of Administrative Reform, 2014
- 1st place in Green Hackathon, Agro-Know, 2013

## Scientific Services

- Reviewer: Association for the Advancement of Artificial Intelligence (AAAI), Artificial Intelligence and Statistics (AISTATS), Transactions of Machine Learning Research (TMLR), Learning on Graphs (LOG), Knowledge Discovery and Data Mining (KDD), European Journal of Operations Research (EJOR), International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), The Web Conference (WWW), Elsevier Expert Systems With Applications, International Conference on Artificial Neural Networks (ICANN), IEEE Advances in Social Networks Analysis and Mining, IEEE Transactions of Intelligent Transportation Systems, IEEE Transactions on Knowledge and Data Engineering, Springer Data Mining and Knowledge Discovery, Springer Artificial Intelligence Review, Springer Scientometrics, Nature Scientific Reports, Springer Applied Network Science, Elsevier Informetrics
- Co-organizing International Conference on BioInformatics and BioEngineering, 2017

## Research Projects

- COVIF (2020): Analysis of COVID19 opinions in social networks.
- ASIP (2019): Translations and merging of medical ontologies.
- ESIGMA (2019):Research graph mining and graph theory.
- SmartLaw (2018): Develop natural language processing for french legal documents.
- Toyota Safety (2017): Research the impact of stress on driving behavior.
- SemaGrow (2016): Develop data management techniques for farmers.
- C2Learn (2014-2015): Develop creativity-boosting applications for children's multi-lingual essays.

## Technical Skills

- Coding: Python (advanced), Java (intermediate), Scala, C++ (basic)
- Machine Learning: PyTorch, PyG (advanced), JAX, Tensorflow (intermediate)
- Statistics: R (advanced), Matlab (intermediate), Julia (basic)
- Databases: MySQL, PostgreSQL (advanced), AWS ETL (intermediate), Spark, MongoDB, Hadoop (basic)
- Optimization: XPRESS (intermediate), CVXPY (basic)

## References

- Jun Pang (supervisor), Michalis Vazirgiannis (ex-supervisor), Fragkiskos Malliaros (ex-supervisor), Georgios Paschos (ex-director), Nikolaos Liakopoulos (ex-manager).